

Human Capital Theory: Applications to Education and Training

ECON 300 - Labour Economics

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Learning Objectives

- Analyze schooling and on-the-job training as standard investments: compare costs and benefits.
- Recognize the **opportunity cost of time** as the largest cost.
- Understand why simple earnings comparisons can mislead (ability bias; signalling/screening).
- Summarize how returns to education in Canada are large and have risen over time.
- Identify when workers versus firms should pay for training.

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Human Capital Theory: Core Ideas

- Investments made to **raise productivity and earnings**.
- Costs today (tuition, materials, **foregone earnings**); benefits tomorrow (higher income).
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- **Real costs** vs. **transfer costs**.

Figure 9.1 — Age-earnings profiles with vs. without further schooling.

Insert figure/diagram here later.

Age–Earnings Profiles

- Increase with age but at a **decreasing rate**.
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Optimal Investment: Assumptions

- No direct utility/disutility from education (consumption value ignored).
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- Income streams by education level are known.
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NPV of Benefits: Setup

Consider an 18-year-old HS graduate choosing work vs. college.

Let working life be $T - 18$ years and constant income Y .

Present value of the income stream at 18:

$$PV = \sum_{t=1}^{T-18} \frac{Y}{(1+r)^t}$$

where r is the market (discount) rate.

Decision Rules: Margins and IRR

Cost of one more year (MC):

$$MC = Y + D$$

Benefit of one more year (MB):

$$MB = \sum_{t=1}^{T-19} \frac{\Delta Y}{(1+r)^t}$$

Optimal years of schooling when **MB = MC**.

Internal rate of return (IRR): choose schooling until $i = r$.

Figure 9.2: Two Ways to State the Decision Rule — *Placeholder*

Figure 9.2 — (1) $MB=MC$ in PV terms; (2) Internal rate of return i equals market r .

Insert figure/diagram here later.

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- Less incentive with **interrupted careers**.
- Progressive taxes can reduce demand for education.
- Student loans alter marginal costs and attainment.

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Worked Example: Computing Returns

Mary (18) can work now at \$30,000 or take **one year** of post-secondary.

Interest rate $r = 5\%$. Work until 65.

After school, income becomes \$33,000.

Approximation: For a level annuity, $PV \approx \frac{Y}{r}$ (infinite horizon approximation).

The extra year raises PV of income by roughly \$23,943.

IRR: $i = \frac{\Delta Y}{Y+D}$. If $D = 0$, $i = 3000/30000 = 10\%$. If $D = 3000$, $i \approx 9\%$.

Direct costs would need to rise to about \$30,000 for Mary not to invest.

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Figure 9.3 — Offered wage vs. signalling cost; separating equilibrium intuition.

Insert figure/diagram here later.

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- Differentials widen with age (wider at 50 than 20–30).

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Figure 9.4: Earnings by Age and Education, Canadian Males, 2015 — Placeholder

Figure 9.4 — Age-earnings profiles by education group, males (2015).

Insert figure/diagram here later.

Table 9.1: Estimates of the Private Returns¹ to Schooling in Canada, 2000

Level of Schooling	Males	Females
Bachelor's degree ²	12	14
Master's degree	3	5
Ph.D.	nc ²	4
Medicine	21	22
Bachelor's Degree by Field of Study	Males	Females
Education	9	14
Humanities and fine arts	nc	10
Social sciences ³	11	14
Commerce	9	19
Natural sciences	9	8
Engineering and applied science	9	14
Health sciences	18	18

Notes: ¹ Private monetary returns. ² "nc" reported where not computed. ³ Includes economics and related fields.

Human Capital Earnings Function

$$\ln Y = \alpha + rS + \beta_1 \text{EXP} + \beta_2 \text{EXP}^2 + \varepsilon$$

- Y : earnings, S : years of schooling (return r), EXP : experience proxy, ε : unobservables.
- Controls help separate schooling effects from experience and ability.

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Figure 9.5: Log Earnings by Years of Schooling (Women 35–39, 2015) — Placeholder

Figure 9.5 — Log earnings vs. years of schooling with fitted line.

Insert figure/diagram here later.

Table 9.2: Estimated Returns to Schooling and Experience, 2015
(Dependent Variable: Log Annual Earnings)

	Men	Women
Intercept	9.232 (714.15)	8.605 (628.29)
Years of schooling	0.080 (105.31)	0.114 (140.77)
Experience	0.054 (75.96)	0.041 (62.51)
Experience squared	−0.0009 (60.14)	−0.0006 (43.85)
R^2	0.136	0.204
Sample size	121,947	99,398

Ability Bias, Signalling, and Screening

- Difficult to fully control for innate ability, motivation, perseverance.
- Natural experiments used to address bias: twins; compulsory schooling laws; proximity to college.
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Figure 9.6 — Profiles of wages/costs under general vs. specific training.

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Training and the Role of Government

- Markets may underprovide training due to imperfect information or constraints.
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- Human capital investment: compare PV benefits with costs; stop when $MB = MC$ or $i = r$.
- Education shifts age–earnings profiles; earnings functions quantify returns.
- Signalling/screening complicate naïve comparisons; empirical strategies address ability bias.
- On-the-job training: who pays depends on portability (general vs. specific).
- Policy: early education, access to finance, and targeted training can improve outcomes.

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Thank You!